Assessing Efficacy of Geo-SAM AI Models for Dynamic Pathing of Disaster Scenes Utilizing Autonomous Search and Rescue Robotics

Introduction

Autonomous ground robots in **disaster environments** face critical navigation challenges due to limited visibility, unpredictable obstacles, and **rapidly changing terrain**. Ground-only systems relying on onboard sensors struggle in cluttered scenes **obstructed by smoke** and debris. Static aerial mapping becomes **outdated within minutes**. Teleoperated systems require constant human attention, cannot scale to **multiple robots**, and will introduce operator fatigue.

Figure 1: Aerial Photograph of Feb 2023 Earthquake in Turkey (UN Maps, 2023)

To address these concerns, a novel **real-time aerial-guided navigation system** that combines algorithms was developed. A **DJI Ryze Tello EDU drone** provides overhead imagery, which is processed by a **Geo-SAM deep learning model** to classify terrain as **safe or unsafe**. Three path planning algorithms **(A*, RRT*, Greedy)** generate routes through the resulting grid map, while a **dynamic replanning system** responds due to environmental changes. This research advances the field of **disaster robotics** by addressing fundamental gaps in **autonomous navigation.**

Purpose

Survival rates in building collapses drop from **90% to 20% within 24 hours**, yet rescuers waste time on safety. **Real-time aerial guidance** and **AI** enable robots to **autonomously navigate** changing situations quickly, **reaching survivors** where humans cannot.

Methods

1. Aerial Imagery

DJI Tello EDU Drone takes aerial imagery of disaster scene

Figure 2: DJI Drone (Photograph taken by Finalist, 2026)

Figure 3: Image 297 from LADI v2 Dataset (LADIv2 dataset, 2024)

2. Segmentation

GeoSAM AI model segments and classifies terrain safety for path planning algorithms

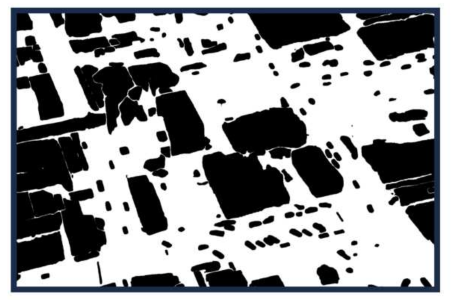

Figure 4: Black and White Binary Mask (Image created by Finalist using Python, 2026)

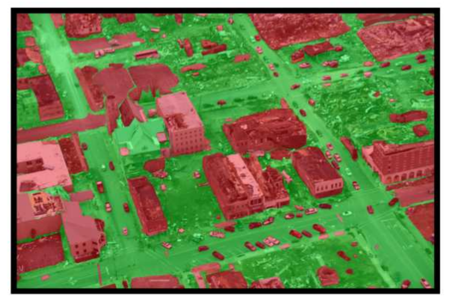

Figure 5: Green and Red Classification Overlay (Image created by Finalist using Python, 2026)

3. Action

Robot navigates using generated instructions from best safe path

Figure 6: Raspberry Pi Robot (Photograph taken by Finalist, 2026)

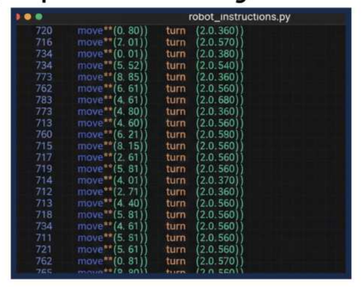

Figure 7: Example of Generated Robot Commands (Photograph taken by Finalist, 2026)

Pathing Algorithms

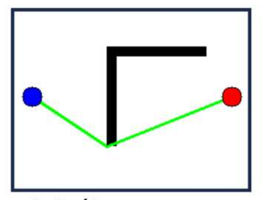

Figure 8: A* (Image created by Finalist using Python, 2026)

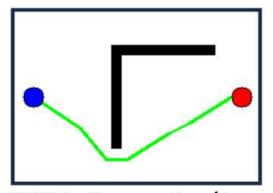

Figure 10: RRT* Run 1 (Image created by Finalist using Python, 2026)

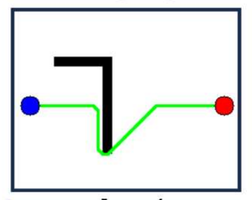

Figure 9: Greedy (Image created by Finalist using Python, 2026)

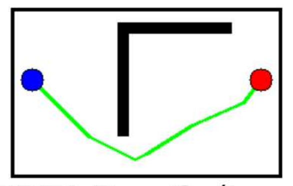

Figure 11: RRT* Run 2 (Image created by Finalist using Python, 2026)

GeoSAM Segmentations

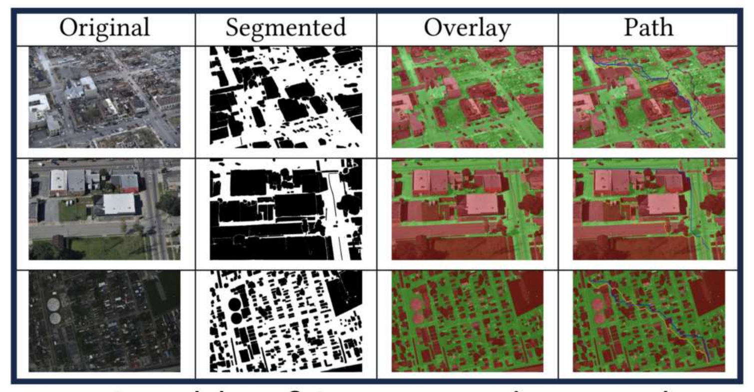

Figure 12: Table of Segmentation results on 3 Different images of the LADI v2 Dataset (Table created by Finalist Using Overleaf, 2026)

Results

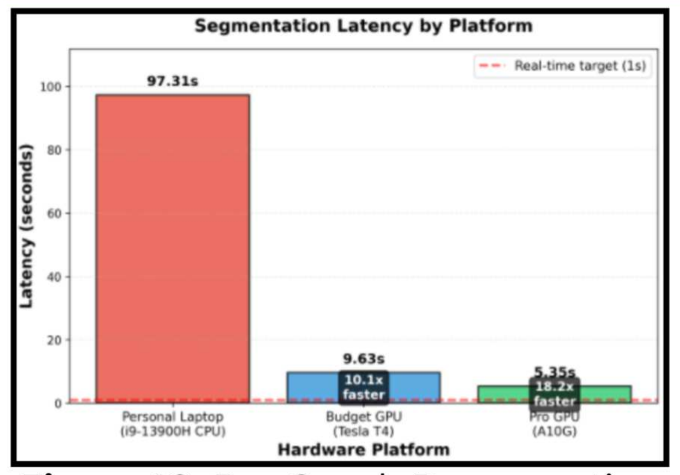

Figure 13: Bar Graph Representing Image Segmentation Times Per GPU/CPU (Personal Laptop - 97.31s, NVIDIA T4 - 9.63s, NVIDIA A10 - 5.35 s) (Graph created by Finalist using Python, 2026)

Figure 14: Scatter Plot Displaying Speed Vs Optimality of Different Algorithms (A*: 148 ms - 1.18x, RRT*: 483 ms - 1.34x, Greedy: 53 ms - 1.47x) (Graph created by Finalist using Python, 2026)

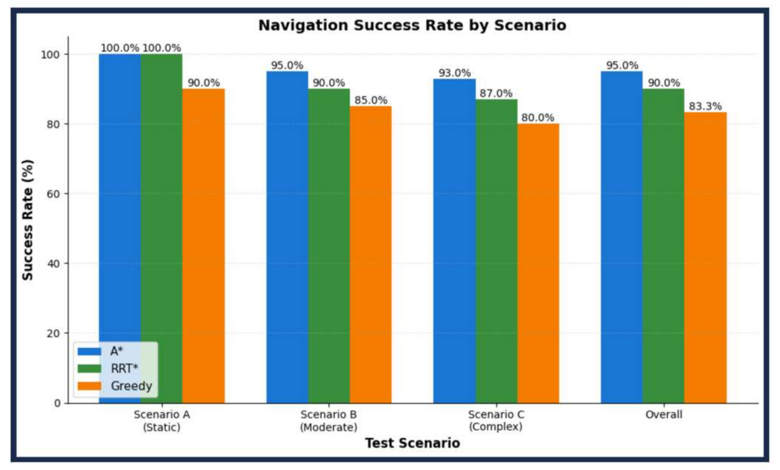

Figure 15: Bar Graph Displaying Navigation Success Rate per Scenario (A*: [100%, 95%, 93%, 95%]; RRT*: [100%, 90%, 87%, 90%]; Greedy: [90%, 85%, 80%, 83.3%]) (Graph created by Finalist using Python, 2026)

Analysis

GPU accelerated SAM segmentation latency by 94.85%, dropping from **97.31s to 5.35s, enabling real time terrain analysis. A*** demonstrated **superior performance**, balancing **path optimality (1.18×)** with **computational efficiency (148ms).** Its **95% success rate** shows **reliability** for disaster navigation. **RRT*** handled **complex spaces** effectively but took **483ms**. **Greedy's 52ms** planning time suits time-critical scenarios despite lower success rate. **Dynamic replanning** adapted to changes in **97% of events**. The **32% improvement** over ground-only systems validates overhead navigation. Limitations include WiFi range (30m), battery life (12 min), and AI performance.

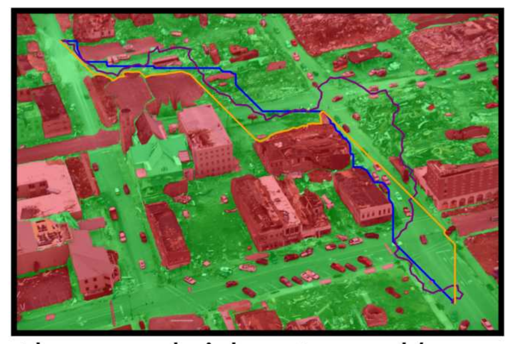

Figure 16: All paths overlaid onto red/green classification. (A*:Blue, RRT*:Purple, Greedy: Yellow) (Image created by Finalist using Python, 2026)

Applications

This system has immediate applications in **disaster response (earthquake rubble, building collapse, hazmat spills)** and **search-and-rescue operations**. Some broader relevant applications include construction site navigation, agricultural robots, warehouse automation, space exploration, and military reconnaissance in GPS-denied environments. Future work will explore **multi-robot swarms** and **deep learning integration.**

References

- Karur, K., Sharma, N., Dharmatti, C., & Siegel, J. E. (2021). A survey of path planning algorithms for mobile robots. Vehicles, 3(3), 448-468. https://doi.org/10.3390/vehicles3030027
- Sultan, R., Li, C., Zhu, H., Khanduri, P., Brocanelli, M., & Zhu, D. (n.d.). (2025) GeoSAM: Fine-tuning SAM with Multi-Modal Prompts for Mobility Infrastructure Segmentation https://arxiv.org/pdf/2311.11319
- Ušinskis, V., Nowicki, M., Dzedzickis, A., & Bučinskas, V. (2025). Sensor-fusion based navigation for autonomous mobile robots. Sensors, 25(4), 1248. https://doi.org/10.3390/s25041248